

1. Tree

1.1 binary tree: definition

1.2 height

1.2.1 definition

1.2.2 recursive form

1.2.3 height of empty tree

1.3 full, perfect, complete

1.4 null pointers

2. Traversal

2.1 preorder

2.2 inorder

inorder=sorted order

2.3 postorder

2.4 lever-order traversal

3. BST

3.1 definition

3.2 the range of height

in this part, pay attention to the big-O notation

3.3 BST implementation

review code here, and pay attention to the running time

3.3.1 BST class definition

3.3.2 find and running time

3.3.3 insert and running time

3.3.4 remove

3.3.4.1 0 children

3.3.4.2 1 child

3.3.4.3 2 children

3.3.4.4 IOP

3.4 runing time

for the balance case, for the average case, and for the worst case

4. AVL Tree

4.1 from BST to AVL

why we need AVL? Review the motivation we use AVL

4.2 balance factor

4.3 definition

4.4 the connection between BST and AVL

4.5 rotation

right, left, right-left, left-right
review code and running time here

4.6 insert

four steps and running time

5. Huffman